Elements of Financial Management

VI Semester SEE Question Papers

Course codes: IM266TEQ, 21IE6F8

- Aug 2025 (IM266TEQ)
- Sep/Oct 2024 (21IE6F8)

IM266TEQ

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

Industrial Engineering and Management

ELEMENTS OF FINANCIAL MANAGEMENT (Institutional Elective)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Interest factor table is permitted.

PART-A

			M	BT	CO
1	1.1	List the three broad areas of Financial Decision making in the organization.	02	1	1
	1.2	What is meant by the term Financial Intermediary? Give an example.	02	2	1
	1.3	List the three important Financial Statements that help in understanding the financial health of a firm.	02	1	1
	1.4	Distinguish between the terms Par Value and Coupon Rate in the context of Bond/Debenture.	02	2	2
	1.5	What is meant by the term "Yield to Maturity"?	02	2	2
	1.6	List the three distinct features of Capital Budgeting.	02	1	3
	1.7	Distinguish between the terms, Authorized Capital and Issued Capital.	02	2	4
	1.8	Why is Preference Capital considered as Hybrid form of Financing.	02	2	3
	1.9	List any two types of Intercorporate Deposits.	02	1	3
	1.10	What is meant by Factoring in the context of Working Capital Financing?	02	2	3

PART-B

2	a	Critically assess the contribution of Financial Intermediaries in the financial inclusion and economic development of a nation.	08	3	1
	b	Examine the recent growth trends in the Indian Financial System. In your answer, focus on to what extent digitization contributed to this growth.	08	3	1
3	a	Explain the key components of a firm's annual report.	05	2	2
	b	A company decides to deposit $Rs.20{,}000$ at the end of each year for 5 years in an account that earns 7% annual interest, compounded annually. Calculate the future value of this annuity.	05	3	2
	c	A firm expects to receive the following cash flows over the next 4 years: $Year$ | $Cash\ Flow\ (Rs)$ 1 | 5,000 2 | 7,000 3 | 6,000 4 | 8,000 If the discount rate is 10%, calculate the present value of these cash flows.	06	3	2

OR

4 a A bond has the following features:

- Par Value = $Rs.1{,}000$
- Coupon Rate = 8% (annual interest)
- Time to Maturity = 5 $years$
- Required Rate of Return (Discount Rate) = 10%

Calculate the value of the bond. 05 3 2

b A bond with a par value of $Rs.1{,}000$, coupon rate of 10% (annual coupon), and 5 years to maturity is currently priced at $Rs.920$. Calculate the approximate Yield To Maturity (YTM) using the formula approach. 05 3 2

c A firm is expected to grow at the rate of 6% per annum. The dividend expected on the firm's equity share a year hence is $Rs.2/-$. What price will you put on it if your Required Rate of Return for this share is 14%? 06 3 2

5 a Distinguish between Market Risk and Unique Risk. Support your explanation with suitable example for each. 06 2 2

b An investor is analyzing a stock that performs differently under various economic conditions. The following data summarizes the possible states of the economy, the probability of each state, and the expected return in each state:

State of Economy	*Probability*	*Rate of Return* (%)
Boom	0.3	20
Normal	0.5	12
Recession	0.2	−5

Calculate:

(i) The expected Rate of Return

(ii) The standard Deviation of Return. 10 3 2

OR

6 a Enumerate the key phases involved in Capital Budgeting Process. 06 2 3

b A project requires an initial investment of $Rs.50{,}000$ and is expected to generate the following cash flows over 4 years: Calculate the Internal Rate of Return (IRR) for the project using interpolation method.

Year	*Cash Flow* (*Rs.*)
1	15,000
2	18,000
3	20,000
4	25,000

10 3 3

7 a The choice of long-term finance depends heavily on the type, size, and stage of a business. Critically evaluate this statement by comparing the suitability of different long-term financing sources for: "A newly established tech startup". 08 4 4

b Highlight the major phases involved in raising long term finance through Term loans. 08 2 4

OR

8	a	What does a Stock Market Index represent? Explain the three important types of indices used.	08	2	4
	b	Explain the structure and features of the Government Securities (G-Sec) Market in India. Why are G-Secs considered low-risk investments, and what role do they play in the financial system?	08	3	4
9	a	"Working capital requirements vary significantly across industries and business models." Critically analyze this statement by identifying and evaluating four key factors that influence working capital needs, with relevant real-world examples to support your analysis.	08	4	3
	b	Explain the three strategies followed by the organizations as part of Current assets financing policy.	08	2	3
		OR			
10	a	"The length of the operating cycle and cash cycle directly affects a firm's liquidity and financing decisions." Evaluate how variations in these cycles impact working capital requirements and business sustainability.	08	3	3
	b	"Efficient working capital financing is critical for maintaining business liquidity without overburdening the firm with cost or risk." In the above context explain any four methods of working capital financing.	08	2	3

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B.E. Regular Examinations Sept/Oct – 2024

ELEMENTS OF FINANCIAL MANAGEMENT (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Usage of interest factor tables is permitted.

PART-A

			M	BT	CO
1	1.1	List the three broad areas of financial decision making.	02	01	01
	1.2	What is the primary function of a financial intermediary in Indian financial system?	02	02	01
	1.3	What is the meaning of "perpetuity" in the context of Time value of money?	02	02	01
	1.4	Distinguish between the terms, "Market Risk" and "Unique Risk".	02	02	01
	1.5	List the 3 distinct features of capital expenditure.	02	01	02
	1.6	What is meant by the term, "Cost of retained earnings"?	02	02	02
	1.7	Distinguish between 'primary Market' and 'Secondary market'.	02	02	03
	1.8	Distinguish between 'Initial Public Offer' and 'Follow on public offer'.	02	02	03
	1.9	Distinguish between the terms – Absorption and Consolidation in the context of Mergers.	02	02	04
	1.10	Distinguish between the terms – call option and put option.	02	02	04

PART-B

2	a	What is meant by a financial market? Summarize the categorization of financial markets based on several criterion.	08	03	01
	b	List any four types of financial intermediaries prevalent in India. Evaluate the benefits of having financial intermediaries in the financial system.	08	03	01
3	a	An amount of $Rs\,1000/-$ is annually deposited for 5 years and the deposit earns a compound interest rate of 10%. What will be the value at the end of 5 years? Assuming that each deposit occurs at the end of the year, determine the future of annuity.	05	03	01
	b	Your friend promises to give you $Rs.\,1000/-$ 6 years later. What is the present value of this amount if the interest rate is 10%?	05	03	01
	c	An investor expects to receive the following cash flows from an asset. *Year* | 1 | 2 | 3 *Cash Flow (Lakhs in rupees)* | 20 | 30 | 220 The discount rate for cash flows is 16%. Determine the value of the Asset.	06	03	01
		OR			
4	a	A $Rs\,1000/-$ par value bond, bearing a coupon rate of 14% will mature after 5 years. The required rate of return on this bond is 13%. What is the value of Bond?	05	03	01

	b	An equity share is expected to provide a dividend of $Rs\,2/-$ and fetch a price of $Rs\,18/-$ a year hence. What price would it sell for now if investors required rate of return is 12%?	05	03	01
	c	The expected dividend per share is Rs.5/-. The dividend is expected to grow at the rate of 6% per year. If the price share now is Rs.50/-, determine the expected rate of returns of the equity share.	06	03	01
5	a	Enumerate the steps involved in the process of capital budgeting.	10	02	02
	b	The cashflow of a project being considered by a company are as follows: *Year* | 0 | 1 | 2 | 3 | 4 *Cashflow (lakhs)* | 1 | 0.3 | 0.3 | 0.4 | 0.45 Calculate the Internal Rate of Returns of the project.	06	03	02
		OR			
6	a	Consider a debenture with a face value of Rs. 1000/-. The coupon rate is 12%. The remaining period to maturity is 4 years. The current market price of the debenture is Rs. 1040/-. Determine the cost of this debenture.	06	03	02
	b	The current market price of an equity is Rs. 80/-. The dividend expected a year hence from now is Rs. 4/-. The growth is expected to be at a constant rate of 10%. The floatation cost for new issue is 8%. Determine the cost of new equity.	05	03	02
	c	A project which is being evaluated by firm has a cost of capital of 12%. The initial investment is Rs. 1 lakh. The benefits over the years are as shown below: *Year* | 1 | 2 | 3 | 4 *Amount (Rs)* | 25,000 | 40,000 | 40,000 | 50,000 Determine the Benefit Cost Ratio of this investment and interpret whether this project is acceptable or not acceptable based on *BCR*?	05	04	02
7	a	Enumerate the five major 'sources' of long term finance available for organizations.	10	02	03
	b	Distinguish between 'Operating Cycle' and 'Cash Cycle' with the help of an illustration.	06	02	03
		OR			
8	a	These are three broad types of 'Indices' used to measure the movement of stocks in a market. Summarize the salient aspects of the three categories/types of indices.	06	02	03
	b	Enumerate the factors that have a crucial bearing on the working capital requirements of an organization.	10	02	03
9	a	Express the plausible reasons for mergers. Analyze how the mergers could contribute to enable organizations have a competitive edge over other organizations.	10	04	04
	b	"The warrant represents a potential source of capital in the future when the organization needs to raise additional capital without offering other bonds or stock". In the light of this statement, determine the features of Warrants.	06	02	04
		OR			
10	a	Examine any four ways of raising the foreign currency finance.	10	04	04
	b	Bring out the rationale of organizations going for 'Demerger' Strategy.	06	03	04

Optimization Techniques

VI Semester SEE Question Papers

Course codes: IM266TER

- Aug 2025 (IM266TER)

IM266TER

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

VI Semester B. E. Regular Examinations August-2025

OPTIMIZATION TECHNIQUES

(Institutional Elective)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of Statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	Operations research is a _____ approach to problem solving for executives.	1	1	1
	1.2	A linear programming technique improves the quality of _____.	1	1	2
	1.3	There are many real world problems that don't possess any _____ solution.	1	1	1
	1.4	If an optimal solution to a linear programming problem exists, it will lie at _____of the feasible solution.	1	2	2
	1.5	Define sensitivity analysis in linear programming.	2	1	2
	1.6	What does the shadow price in duality represent?	2	2	2
	1.7	Explain degeneracy in transportation problems.	2	2	2
	1.8	State the objective of the Traveling Salesman Problem.	2	2	2
	1.9	List the three time estimates used in PERT.	2	1	3
	1.10	What are the advantages of crashing a network?	2	1	3
	1.11	Differentiate between pure and mixed strategies.	2	1	3
	1.12	Explain the characteristics of a two-person zero-sum game.	2	1	3

PART-B

2	a	A company produces two products A and B. Each unit of A takes 2hrs of machine time and 3 units of labor. B takes 4hrs of machine time and 2 units of labor. The machine and labor availability are 100hrs and 90 units. The profit from A is ₹30 and from B is ₹50. Formulate the LPP.	06	3	2
	b	Explain in detail the seven-step model-building process with a real-life example for each step.	10	2	1
3	a	Solve the following LP using the Simplex Method: Maximize $Z = 4X_1 + 3X_2$ $subject\ to$: $2X_1 + X_2 \leq 10$, $X_1 + 2X_2 \leq 6$ $X_1, X_2 \geq 0$	10	3	2
	b	With suitable example, explain slack and surplus variables.	06	2	2
		OR			

4	a	Formulate the dual of the following problem and interpret the dual variables economically: Maximize $Z = 6X_1 + 8X_2$ $Subjected\ to$: $2X_1 + 3X_2 \leq 30$, $X_1 + 2X_2 \leq 20$, $X_1, X_2 \geq 0$	06	4	2
	b	Use the Simplex Method to solve the LP: Maximize $Z = 3x_1 + 2x_2$, subject to $x_1 + x_2 \leq 4$, $x_1 + 2x_2 \leq 6$, $x_1, x_2 \geq 0$	10	3	2
5	a	A manufacturer has 2 plants (P1 and P2) with production capacities of 100 and 150 units. These must be shipped to 3 warehouses (W1, W2, W3) requiring 80, 120, and 50 units, respectively. The cost of transportation per unit is as follows: $W1$ $W2$ $W3$ $P1$ 4 6 8 $P2$ 5 3 7 Formulate the transportation problem mathematically.	08	3	2
	b	In a textile sales emporium, four salesmen A, B, C and D are available to four counters W, X, Y and Z. Each salesman can handle any counter. The service (in hour) of each counter when manned by each salesman is given below: $Salesman$: A B C D $Counter$: W 41 72 39 52; X 22 29 49 65; Y 27 39 60 51; Z 45 50 48 52 How should the salesmen be allocated appropriate counters so as to minimize the service time? Each salesman must handle only one counter.	08	3	2
		OR			
6	a	Use the following data to find IBFS using: (i) North-West Corner Rule (NWCR) (ii) Least Cost Method (LCM) $D1$ $D2$ $D3$ $Supply$ $S1$ 2 3 1 30 $S2$ 5 4 8 50 $S3$ 5 6 8 20 $Demand$ 20 40 40	08	3	2
	b	A car rental company has one car at each of five depots I, II, III, IV and V. A customer requires a car in each town namely A, B, C, D and E. The distance (in km) between depots and towns are given in the following distance matrix. $Depots$: I II III IV V $Town$: A 160 130 175 190 200; B 135 120 130 160 175; C 140 110 155 170 185; D 50 50 80 80 110; E 55 35 70 80 105 How should the cars be assigned to the customers so as to minimize the distance travelled?	08	3	2

7 An established company has decided to add a new product to its line. It will buy the product from a manufacturing concern, package it, and sell it to a number of distributors that have been selected on a geographical basis. Market research has already indicated the volume expected and the size of sales force required. The steps shown in the following table are to be planned.

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M
Duration	6	4	7	2	4	10	2	10	6	13	9	3	5
Predecessors	-	A	B	A	D	E	-	G	J,H	-	A	C,K	I,L

(i) Draw an arrow diagram for this project.
(ii) Indicate the critical path.
(iii) For each non-critical activity, find the total and free float — 16 | 4 | 3

OR

8 A small project involves 7 activities, and their time estimates are listed in the following table. Activities are identified by their beginning (i) and ending (j) node numbers.

Activity ($i-j$)	*Estimated Duration (weeks)*		
	Optimistic	*Most Likely*	*Pessimistic*
1 – 2	1	1	7
1 – 3	1	4	7
1 – 4	2	2	8
2 – 5	1	1	1
3 – 5	2	5	14
4 – 6	2	5	8
5 – 6	3	6	15

(i) Draw the network diagram of the activities in the project.
(ii) Find the expected duration and variance for each activity. What is the expected project length?
(iii) Calculate the variance and standard deviation of the project length. What is probability that the project will be completed:
I. at least 4 weeks earlier than expected time.
II. no more than 4 weeks later than expected time. — 16 | 4 | 3

9 a Use the graphical method to find the optimal strategy for both the players and also value of the game for the following pay – off matrix:

		Player B		
		B1	*B2*	*B3*
Player A	*A1*	1	3	12
	A2	8	6	2

10 | 3 | 4

b Explain: Minimax and Maximin principle used in the theory of games. 06 | 2 | 4

OR

10 a Use graphical method to find the optimal strategy for both the players and also value of the game for the following pay – off matrix:

		Player B				
		I	*II*	*III*	*IV*	*V*
Player A	*I*	3	0	6	–1	7
	II	–1	5	–2	2	1

10 | 3 | 4

b State the major limitations of the game theory. 06 | 2 | 4

Mathematical Modelling

VI Semester SEE Question Papers

Course codes: MA266TEU

- Jul/Aug 2026 (MA266TEU)
- Model Question Paper (2022 Scheme) (MA266TEU)

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B.E. Regular / Supplementary Examination July / August 2026

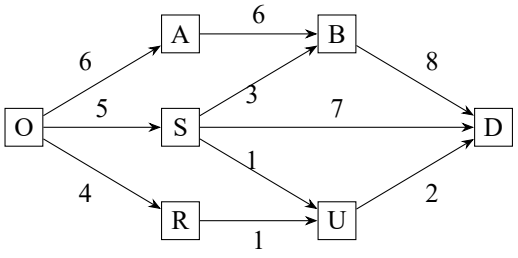
(Institutional Elective)

MATHEMATICAL MODELLING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

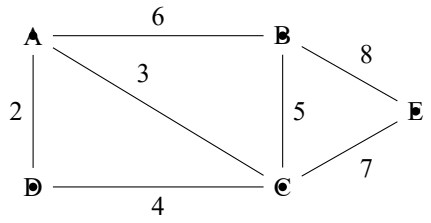
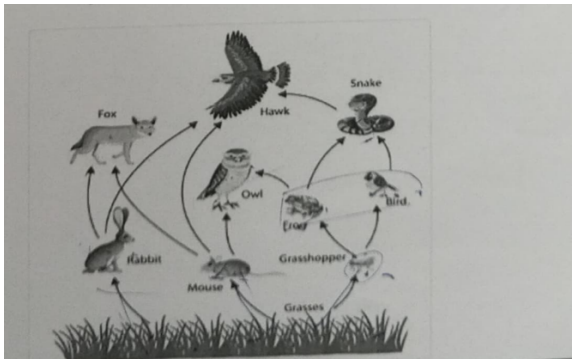
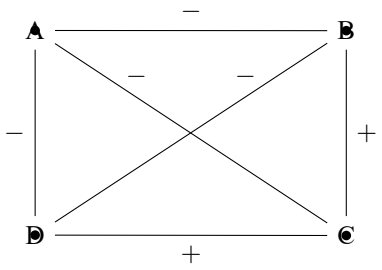
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Write any two characteristics of mathematical modelling.	02	1	1
	1.2	The horizontal distance (range) covered by a soccer ball of mass 2 kg is kicked from the ground with an initial velocity of $30\,m/s$ at an angle of 30° to the horizontal is _____. (Take acceleration due to gravity as $g \approx 9.8\,m/s^2$)	02	2	2
	1.3	With respect to the characteristic of brown eyes, suppose the proportions of dominant, hybrid, and recessive individuals in the 9th generation are $p_9 = 0.36$, $q_9 = 0.48$, $r_9 = 0.16$. The proportion of recessives in the 10th generation (r_{10}) is _____ and proportion of hybrids in the 11th generation (q_{11}) is _____.	02	1	1
	1.4	Suppose the population of one-horned rhinoceros in foothills of Himalayas are decreasing at the rate of 2% per year and their population was 400 in 2015, then how many of them will be left approximately in 2024?	02	2	2
	1.5	Compute the fixed probability vector of the stochastic matrix $A = \begin{bmatrix} 0.5 & 0.5 \\ 0.1 & 0.9 \end{bmatrix}$.	02	1	1
	1.6	A travelling salesman **S** sells in 3 cities **A, B, C**. He never sells in the same city on successive days. If he sells in city A on a day, then the next day he sells in city B. However if he sells in either B or C, then the next day he is twice as likely to sell in city A as in the other city. Find the transition matrix of the chain of selling.	02	2	2
	1.7	Draw the bipartite graph $K_{3,5}$ and the complete graph K_4.	02	1	1
	1.8	Express the following transition probability matrix as a weighted digraph. $\begin{bmatrix} 0.3 & 0.2 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.25 & 0.35 & 0.4 \end{bmatrix}$	02	2	2
	1.9	In the following figure, with respect to dynamic programming, the value of $f_1^*(O)$ is _____. 	02	2	2

	1.10	The extremal of the functional $\int_0^3 x^3(y')^2\,dx$ is _____.	02	1	1

PART-B

2	a	Model and describe the second-order differential equations representing a mass-spring-dashpot mechanical system. Also highlight the analogies between the elements of the mechanical and electrical systems.	08	3	3
	b	Drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing $c(t)$, solve it and interpret the results.	08	4	4
3	a	Discuss the Cobweb Model describing the variation of price and quantity of an agricultural commodity produced by farmers. Using a suitable graph, interpret the behaviour of the model.	08	3	3
	b	(i) A machine's value changes every year according to the relation $V_{n+1} = 0.80V_n - 2$, where V_n is the value in lakhs after n years. If the initial value $V_0 = 15$ lakh, find the value of the machine after n years. (ii) Solve the difference equation $2x_{t+2} - 5x_{t+1} + 2x_t = 0$.	08	2	2
		OR			
4	a	Model the population dynamics using difference equations and discuss the one period fixed points.	08	3	3
	b	Solve the difference equation $x_{t+2} = 4x_{t+1} - 5x_t$.	08	2	2
5	a	An e-commerce website tracks user navigation behaviour between four main sections: The Homepage (H), Product Catalog (C), Shopping Cart (W) and the Payment Gateway (P). From the Homepage (H): Users always navigate directly to the Product Catalog (C). From the Product Catalog (C): There is a 50% chance they stay browsing the Catalog (C) and a 50% chance they add an item to the Shopping Cart (W). From the Shopping Cart (W): There is a 60% chance they return to the Homepage (H), a 20% chance they proceed to the Payment Gateway (P) and a 20% chance they remain in the Cart adjusting items. From the Payment Gateway (P): After completing or cancelling a transaction, they are always redirected back to the Homepage (H). In the long run, what percentage of a user's active session time is spent on the Payment Gateway (P)?	08	4	4
	b	A company executive changes his bike every year. If he has Pulsar he changes to Splendor, if he has Splendor he changes over to Royal Enfield. If he has Royal Enfield, he is just as likely to change over to a bike Royal Enfield, Splendor or Pulsar. If he had Royal Enfield in 2020, find the probability that he will have a bike (i) Pulsar in 2022, (ii) Royal Enfield in 2022.	08	2	2
		OR			
6	a	Verify the transition probability matrix $P = \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 0 \end{bmatrix}$ is irreducible or not. Also find its stationary probability vector.	08	2	2

	b	An automated robotic arm on a factory floor moves items between four inspection stations: Receiving (R), Sorting (S), Processing (P) and Re-inspection (M). From Receiving (R): Items always move directly to Sorting (S). From Sorting (S): There is a 40% chance an item stays in Sorting for deeper scanning and a 60% chance it moves to Processing (P). From Processing (P): There is a 70% chance it moves back to Receiving (R), a 10% chance it requires Re-inspection (M) and a 20% chance it stays in Processing (P). From Re-inspection (M): Once checked, it always moves back to Receiving (R). In the long run, what percentage of time does the robotic arm spend in Re-inspection (M)?	08	4	4
7	a	State the max-flow min-cut theorem. For the network shown below, determine the maximum flow between A and E by enumerating all the A-E cuts and the capacities therein. 	08	4	4
	b	Define the Harary measure of status. In the given food web, an edge from X to Y means X is food for Y. Find the status of each species according to the Harary measure. 	08	3	3
		OR			
8	a	Alvin (A) has invited three married couples to his summer house for a week: Bob (B) and Carrie (C) Hanson, David (D) and Edith (E) Irwin, and Frank (F) and Gena (G) Jackson. Since all six guests enjoy playing tennis match against every other guest except his/her spouse. In addition, Alvin is to play a match against each of David, Edith, Frank, and Gena. If no one is to play two matches on the same day, what is a schedule of matches over the smallest number of days.	08	4	4
	b	i. Is it possible to have a graph with the degree sequence 1,2,2,3,4,5? If yes, draw an example. If not, give reason. ii. Define the degree of balance of a signed graph. The following graph models the social relation between 4 people with '+' indicating friendship and '−' indicating enmity. Find the degree of balance of the graph. 	08	3	3

9	a	Find the extremals of the functional $\int_{1/10}^{1} y'(1+x^2y')\,dx$, if $y\left(\frac{1}{10}\right) = 19$, $y(1) = 1$.	08	3	3
	b	A company needs to transport goods from its main warehouse (location A) to its retail outlet (location J). Along the way, the goods must pass through a series of distribution centres (B, C, D, E, F, G, H, I). At each segment of the journey, transport costs are incurred. The company wants to minimize the total transportation cost from A to J. Using dynamic programming, determine the minimum transportation cost from A to J and identify the optimal route. The transportation costs between centres are given in the table below: A: B 5, C 3, D 6 B: E 4, F 2, G 5 C: E 3, F 4, G 6 D: E 5, F 3, G 2 E: H 3, I 4 F: H 2, I 5 G: H 6, I 3 H: J 4 I: J 3	08	2	2
		OR			
10	a	Model the problem of a uniform chain (cable or rope) of length $2l$, suspended under gravity between two points at equal height. Find the curve when the chain is in equilibrium.	08	4	4
	b	Discuss any five optimization techniques and any three optimization principles.	08	2	2

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

VI Semester B.E. Examinations – Model Question Paper

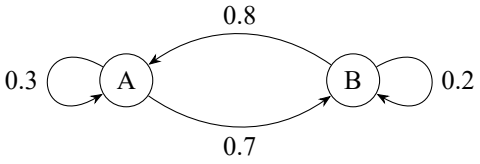
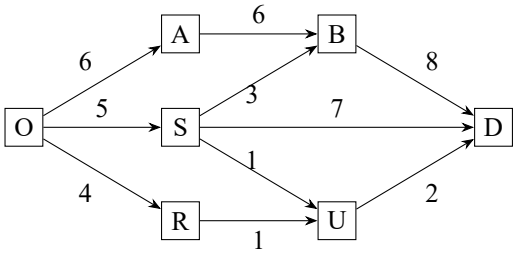
(Institutional Elective)

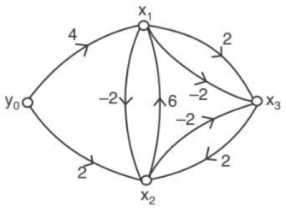
MATHEMATICAL MODELLING

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

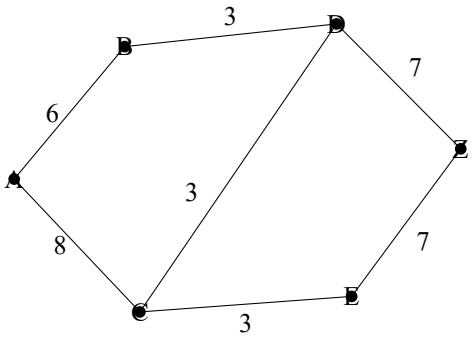
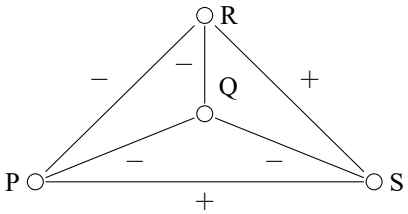
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

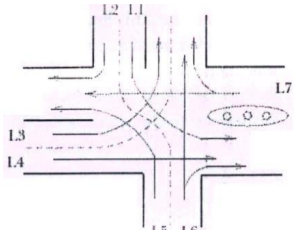
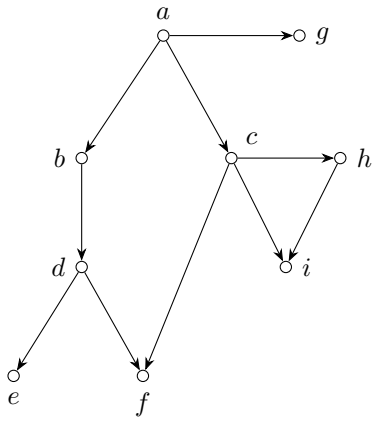
		PART-A	M	BT	CO
1	1.1	The population of California fox are decreasing at the rate of 5% per year. If their population was 400 in 2013, then how many of them will be left approximately in 2023?	02	2	1
	1.2	The growth of an organism follows the law $\frac{dx}{dt} = -\frac{1}{13}x$. If initially there were 200 of them, then after 13 years, their approximate number will be _____ and their half-life period is approximately _____ years.	02	2	3
	1.3	In terms of y_{t+1} and y_{t+2}, the arbitrary constant M in the equation $y_t = M2^t + N(-3)^t$ is equal to _____.	02	3	3
	1.4	The curve which extremizes the functional $\int_{10}^{11} \left[y^2 + x^2 y'\right] dx$ is _____.	02	2	2
	1.5	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is _____. 	02	2	2
	1.6	The angle of projection for a projectile to attain a) the maximum horizontal distance should be _____. b) the maximum height should be _____.	02	1	3
	1.7	In the following figure, with respect to dynamic programming, the value of $f_1^*(O)$ is _____. 	02	2	4

	1.8	Write the linear equations associated with the following signal-flow graph:  	02	2	4
	1.9	Given the initial state vector $(1, 0)$ and the transition matrix shown below, find the state vector corresponding to two steps later ($n = 2$). $\begin{bmatrix} 0.13 & 0.87 \\ 0.91 & 0.09 \end{bmatrix}$	02	1	1
	1.10	'Edge coloring' is the assigning of colors to the edges so that no two adjacent edges have same color. Consider a simple connected graph G in which the number of vertices is equal to the number of edges, and every vertex has degree 2. What is the minimum number of colors required to 'edge-color' G?	02	1	2

PART-B

2	a	Give a note on the classification of mathematical models giving at least four examples.	08	1	1
	b	Drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration $c(t)$ of the drug present at any time t. Determine the differential equation governing $c(t)$, solve it and interpret the results.	08	3	3
3	a	Mathematically establish the Hardy-Weinberg law in genetics.	08	3	2
	b	A person takes a loan of ₹S_0 with i as the compound interest rate per month. If the loan is to be paid back in n equal instalments by paying ₹R per month, then show that $R = S_0 \dfrac{i}{1-(1+i)^{-n}}$.	08	2	2
		OR			
4	a	Bring out the dynamics of price and quantity of a commodity produced by a farm according to the Cobweb model.	08	3	2
	b	i. Solve the difference equation $x_{t+2} = 4(x_{t+1} - x_t)$, $t \geq 0$, $x_0 = 1$, $x_1 = 3$. ii. The number of virus affected files in a system is 5000 (to start with) and this increases 200% every two hours. Model this using a difference equation and find the number of virus affected files in the system after one day.	08	2	2
5	a	Check if the transition probability matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$ is irreducible. Also find its stationary probability vector.	08	2	1
	b	A city is served by two cable TV companies, Best TV and Cable Cast. Due to their aggressive sales tactics, each year 40% of Best TV customers switch to Cable Cast. On the other hand, 30% of the Cable Cast customers switch to Best TV. If the initial market share for Best TV is 20% and for Cable Cast is 80%, find the market share of each company after 5 years.	08	2	3
		OR			

6	a	Consider a network containing five individuals viz. a, b, c, d and e. The system a can communicate to only b and c and its chances of communicating with each are in the ratio 3:2 respectively. Similarly, b communicates with a and c in ratio 1:1; c can communicate with a, b and d in ratio 4:3:3; d communicates with c and e in ratio 3:7 and e can communicate to b only. Model the situation as a Markov chain and rank the five individuals in accordance with their importance in the network in the long run.	08	2	1
	b	A gambler has ₹2. He bets ₹1 at a time and wins ₹1 with probability $\frac{1}{2}$. He stops playing if he ends up with no money or wins ₹4. a. What is the TPM of the related Markov chain? b. What is the probability that he has lost his money at the end of 5 plays? What is the probability that the game lasts more than 7 plays?	08	3	3
7	a	At the end of the school day, all the students plan on driving from school (vertex A) to the concert (at vertex Z). The edges in the graph below represent one-way roads, and the weight of each edge represents the number of vehicles (in hundreds) that particular road can handle in one hour. What is the greatest number of vehicles that can get from school to the concert in one hour? How many vehicles should take each road?  	08	2	4
	b	i. Basavaraj has picked 8 friends for a project team such that they are all friends among themselves. Considering A and B being friends as a single friendship, is it possible that there are totally 38 friendships? Substantiate your answer. ii. Define the degree of balance of a signed graph. The following graph models the social relation between 4 people with '+' indicating friendship and '−' indicating enmity. Find the degree of balance of the graph.  	08	3	4
OR					

8	a	The Fig 8b shows an intersection of two streets where there is often heavy traffic. There are seven traffic lanes $L1, L2, \ldots L7$ where vehicles can enter the intersection of these two streets. A traffic light is located at this intersection. During a certain phase of this traffic light, those cars in lanes for which the light is green may proceed safely through the intersection. Model this situation by a graph giving out the details as to what the vertices and the edges represent. Use that graph to determine whether it is possible with four phases, for cars in all lanes to proceed safely through the intersection.  Fig 8b	08	2	4
	b	Define the Harary measure of status. Find the Harary measure of each individual in the organizational graph shown in Fig 9c.  Fig 9c	08	3	4
9	a	A person decides to travel from a place A to a place J. The journey would require traveling by a stagecoach through different cities on the paths of which there is danger of attack by thieves. Life insurance policies were offered to the passengers, the safest route would be the one with the cheapest total life insurance policy. The details of the insurance from one place to another are given below. Model it through a directed graph and find the route which minimizes the cost using dynamic programming. B C D A 1 3 6 E F G B 5 4 6 C 4 3 2 D 1 4 5 H I E 5 2 F 3 6 G 3 3 J H 5 I 2	08	1	2
	b	Model the problem of light passing from a point A to a point B as a variational problem by applying Fermat's principle of least time. Show that a consequence of this modelling is the Snell's law of refraction.	08	3	4
		OR			
10	a	Discuss any five optimization techniques and any three optimization principles.	08	1	2
	b	A particle descends from one point to another point in the absence of friction under the action of gravity. Model the situation as a variational problem so as to find the path of the quickest descent.	08	3	4